



MODULE 5 OF 5 · PRINTABLE NOTES

Academic Writing and Deep Research Assistance

Synthesise research information, develop structured academic arguments and improve the clarity, coherence and organisation of academic writing with AI as a responsible drafting partner.

AI Disclosure

Citation Styles

Deep Research

Verify Before You Cite

🎯 LEARNING OBJECTIVES

- Use AI deep-research features responsibly: scope, review the trail, and verify before use.
- Apply APA and IEEE citation styles correctly in-text and in the reference list.
- Draft a compliant AI-disclosure statement for a manuscript.
- Spot fabricated citations, invented statistics and unsupported causal claims in an AI-drafted paragraph.
- Convert a reference list between citation styles without losing or altering any detail.

✓ BEFORE YOU START

- Bring a rough draft, outline or the key findings/visuals from Modules 1–4 that you want to write up.
- Decide which citation style your target journal or institution requires (e.g., APA or IEEE).
- Have a free account ready for at least one citation manager (Zotero is free and works well for beginners).



KEY CONCEPTS

Glossary of terms

Terms you will meet throughout this module, defined in plain language.

DOI

Digital Object Identifier, a permanent link to a specific published work.

Plagiarism checker

A tool that compares your text against published sources to flag unoriginal content.

Deep research

An AI feature that searches, reads and synthesises multiple sources into one report.

AI disclosure

A statement in a manuscript declaring where and how AI tools were used.



QUICK REFERENCE

Citation styles at a glance

APA — APA 7th Edition

In-text: (Author, Year). Reference: Author, A. A. (Year). Title of article. Journal Name, Volume(Issue), pages.
<https://doi.org/xxxx>

IEEE — IEEE Style

In-text: [1]. Reference: [1] A. Author, "Title of article," Journal Name, vol. X, no. X, pp. XX–XX, Year.



Trainer's Tip, Dr. Shahizan: AI can tighten a sentence, but it cannot tell you if your argument holds up. Read your AI-assisted draft out loud once before submission. If a paragraph does not make sense when spoken, it usually is not ready yet.



AI TOOLS FOR THIS MODULE

Citation managers and writing assistants

TOOL	CATEGORY	WHAT IT DOES	WHERE TO ACCESS
Zotero	Citation management	Free tool for collecting, organising, citing and sharing research sources.	zotero.org
Mendeley	Citation management	Reference manager and academic network for research, collaboration and discovery.	mendeley.com
EndNote	Citation management	Reference-management software for organising literature and producing bibliographies.	endnote.com
Grammarly	Grammar & editing	Improves grammar, spelling, punctuation, clarity and tone.	grammarly.com
Paperpal	Academic language	Suggestions for grammar, vocabulary, translation, tone and consistency.	paperpal.com
Jenni AI	Academic writing	Assists with drafting, editing, proofreading and locating potential citations.	jenni.ai
Turnitin	Similarity & AI-writing check	Institutional similarity checker reviewing manuscripts for matched text and AI-generated content.	turnitin.com



HOW TO USE

Step-by-step tool guides



STEP-BY-STEP GUIDE

Using AI Deep-Research Features Responsibly

A deep-research feature searches, reads and synthesises many sources into one report. It accelerates the literature-review stage of writing, but it still summarises rather than verifies.

01

Scope the question

Give the tool one focused question, field and time boundary rather than an open-ended topic.

02

Review the trail

Open the sources the tool cites before trusting its synthesis; a deep-research report still summarises, not verifies.

03

Verify before use

Cross-check key claims, statistics and quotations against the original publications before citing them.



STEP-BY-STEP GUIDE

AI Disclosure Builder

Many journals and institutions now require you to declare AI assistance in the manuscript. Turn the tools you actually used into a compliant disclosure statement.

01

List your tools

Name every AI tool used and what each one was used for (e.g., clarity, related-literature search).

02

Generate

Assemble the disclosure statement using the pattern below.

03

Confirm wording

Check your target journal or institution's exact required wording or placement before submission.

✧ DISCLOSURE STATEMENT PATTERN

During the preparation of this manuscript, the author(s) used [tool name] to improve the clarity and readability of the text, and [tool name] to assist in identifying related literature. All AI-suggested content was reviewed, verified against original sources, and edited by the author(s), who take full responsibility for the content of this publication.



HANDS-ON PRACTICE

All exercises, with model answers

Every exercise below matches the hands-on activity on the module's web page. Attempt each one yourself first, then compare against the model answer.

EXERCISE 5.1

~10 min

Using AI deep-research features responsibly

Deep-research tools search, read and synthesise many sources into one report — a fast way to survey a field, but they still summarise rather than verify.

- ☐ Scope: give the tool one focused question, field and time boundary rather than an open-ended topic.
- ☐ Review the trail: open the sources the tool cites before trusting its synthesis.
- ☐ Verify: cross-check key claims, statistics and quotations against the original publications before citing them.

✓ MODEL ANSWER — SAMPLE SCENARIO

- Scoped question: "What deep-learning approaches have been applied to Malaysian palm oil yield prediction between 2019 and 2024?" — focused by field, method and time boundary.
- Report returned: a synthesis citing 9 papers across three approach families (CNN-based, LSTM-based, and hybrid ensemble models).
- Trail reviewed: opened the 3 most-cited sources directly; confirmed each paper's dataset and reported accuracy matched what the deep-research report claimed.
- One overstated claim caught: the report described one paper's result as "state-of-the-art", but the paper itself only claimed a 2% improvement over a single baseline — the wording was corrected before use.

EXERCISE 5.2

~10 min

Draft your own AI disclosure statement

Turn the disclosure pattern into a real statement for your own manuscript. Fill in the tools you actually used and what you used each one for.

- ☐ Name Tool 1 and its purpose (required).
- ☐ Optionally, name Tool 2 and its purpose.
- ☐ Generate the statement and check it reads accurately for what you actually did.
- ☐ Confirm your target journal or institution's exact required wording or placement before submission — this is a starting draft, not a guaranteed match to every journal's policy.

✓ MODEL ANSWER — FILLED EXAMPLE

Tool 1: Paperpal — purpose: improve clarity and academic tone.

Tool 2: Gemini Notebook — purpose: assist in identifying related literature from uploaded sources.

Generated statement: "During the preparation of this manuscript, the author(s) used Paperpal to improve the clarity and readability of the text, and Gemini Notebook to assist in identifying related literature. All AI-suggested content was reviewed, verified against original sources, and edited by the author(s), who take full responsibility for the content of this publication."

EXERCISE 5.3

~15 min

Verify an AI-drafted paragraph

Read the AI-drafted paragraph below as if it were handed to you to include in a manuscript. Before you look at the answer, decide what you would need to check.

- ☐ Does each citation actually exist, and does the source really say what is claimed?
- ☐ Is each statistic traceable to a specific, checkable source, or does it just sound precise?
- ☐ Does any sentence claim causation ("led to", "resulted in") when the evidence only supports correlation?
- ☐ Is there a sweeping generalisation ("universities worldwide", "all students") that is not actually supported?

¶ SAMPLE AI-DRAFTED PARAGRAPH

Artificial intelligence tools have become widely adopted in higher education, with recent studies showing that over 90% of postgraduate students now use generative AI for their coursework (Chen & Patel, 2024). This dramatic increase has been linked to significant improvements in academic performance, with AI-assisted students scoring on average 30% higher on written assignments (Okafor, 2023). As a result, universities worldwide have begun mandating AI literacy training for all incoming students.

✓ REVEAL WHAT IS WRONG WITH THIS PARAGRAPH

- "Chen & Patel, 2024" and "Okafor, 2023" read like plausible academic citations but are fabricated for this exercise — exactly the kind of citation an AI tool can invent. Always search for a citation before trusting it.
- "over 90% of postgraduate students" is a suspiciously round, high figure attributed to unnamed "recent studies" with no traceable source — a classic AI overgeneralisation.
- "scoring on average 30% higher" implies AI use caused better scores, when at most the invented source could show a correlation. Causation needs a controlled study design, not just an association.
- "universities worldwide have begun mandating" is a sweeping claim with no source at all — the kind of confident-sounding filler AI text adds to make a paragraph feel authoritative.

EXERCISE 5.4

~10 min

Convert a reference list between citation styles

Reformatting a reference list by hand is slow and error-prone. Use AI to do the mechanical conversion, then verify every entry against the citation-styles quick reference.

- ☐ Take 3 to 5 references from your own reference list, in whatever style they are currently in.
- ☐ Use the ready-to-use prompt below to convert them to your target style (e.g., APA to IEEE, or the reverse).
- ☐ Check every converted reference against the quick reference table: author order, punctuation, italics and the overall structure.
- ☐ Confirm no reference was dropped, merged, or had a detail (year, volume, page numbers) changed or invented.

✓ MODEL ANSWER — SAMPLE APA → IEEE CONVERSION

Source (APA): Tan, M. Y., & Lim, S. H. (2022). Explainable AI for medical image diagnosis: A systematic review. *Journal of Medical Informatics*, 14(3), 210–229. <https://doi.org/10.xxxx/jmi.2022.0143>

Converted (IEEE): [1] M. Y. Tan and S. H. Lim, "Explainable AI for medical image diagnosis: A systematic review," *Journal of Medical Informatics*, vol. 14, no. 3, pp. 210–229, 2022.

Verification: author order preserved (Tan then Lim), volume/issue/pages carried over exactly, DOI omitted per this journal's IEEE reference style (confirmed against the quick-reference table) rather than dropped by the AI without explanation.

WHY USE AI HERE



Traditional vs AI-assisted approach

Task	Traditional Approach	AI-Assisted Approach
Improving clarity and grammar	Manual proofreading and peer review only	Grammarly or Paperpal suggest edits in real time
Managing citations	Format each reference by hand	Zotero, Mendeley or EndNote auto-format citations and bibliographies
Finding related work while writing	Pause writing to search separately	Deep-research tools surface related literature inline, with sources cited
Reviewing a manuscript before submission	Wait for a colleague or supervisor's read-through	Manuscript Review Prompts generate a structured editorial report instantly

The trade-off: AI speeds up polish and formatting; your argument, voice and judgement stay firmly in your own hands.

COMMON PITFALLS



Things to verify before you trust the output

- ☐ Submitting an AI-drafted paragraph without verifying every citation actually exists and says what is claimed.
- ☐ Letting a correlational finding be reworded into causal language ("led to", "resulted in") during AI-assisted drafting.
- ☐ Skipping the AI-disclosure statement, or using generic wording that does not match your target journal's exact policy.
- ☐ Trusting a deep-research synthesis without opening at least the top few cited sources yourself.
- ☐ Assuming an AI-converted reference list preserved every detail – always check author order, volume, issue and page numbers against the original.

✦ Key Takeaways

- Use AI to polish clarity and structure, but keep your own argument, voice and judgement in control.
- Verify every fact, statistic, quotation and citation an AI drafted or paraphrased. Never assume it is correct.
- Disclose AI use where your journal or institution requires it.
- A citation manager (Zotero/Mendeley/EndNote) removes formatting errors so you can focus on the argument.

- **What's next:** You've gone from literature discovery to academic writing. Revisit any module's hands-on tool anytime, or see the full-day schedule for breaks and the closing wrap-up.



FOR YOUR NOTES

Session Notes

A large rectangular area with horizontal lines for taking notes.



FOR YOUR NOTES

Session Notes (continued)

A large rectangular area with horizontal lines for taking notes.